Functions: Composite, Inverse & Logarithmic

ANSWER KEY

Section 1: Inverses & Domain Restrictions

1 If $f(x) = \frac{2x-5}{3x+1}$, what is the value of $f^{-1}(1)$?

a. -6 ✓

b. -5

c. -4

d. -2

Set $f(x) = 1$: $\frac{2x-5}{3x+1} = 1 \Rightarrow 2x - 5 = 3x + 1 \Rightarrow -x = 6 \Rightarrow x = -6$.

2 What is the domain of $f^{-1}(x)$ if $f(x) = \sqrt{3 - 2x}$?

a. $[0, \infty)$ ✓

b. $(-\infty, 1.5]$

c. $[1.5, \infty)$

d. All real numbers

The domain of f^{-1} is the range of f. Since $f(x) = \sqrt{3 - 2x} \geq 0$, the range of f is $[0, \infty)$.

3 Given $f(x) = x^2 - 4x + 7$ defined on the restricted domain $[2, \infty)$, what is the equation of $f^{-1}(x)$?

a. $f^{-1}(x) = 2 + \sqrt{x - 3}$ ✓

b. $f^{-1}(x) = 2 - \sqrt{x - 3}$

c. $f^{-1}(x) = \sqrt{x - 7} + 2$

d. $f^{-1}(x) = 2 + \sqrt{x + 3}$

Complete the square: $y = (x-2)^2 + 3 \Rightarrow y - 3 = (x-2)^2$. For $x \geq 2$: $x - 2 = \sqrt{y - 3} \Rightarrow x = 2 + \sqrt{y - 3}$. Thus $f^{-1}(x) = 2 + \sqrt{x - 3}$.

Section 2: Composition & Decomposition of Functions

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4 If $f(x) = 3x - 2$ and $(f \circ g)(x) = 6x^2 + 15x + 4$, what is the expression for $g(x)$?

a. $2x^2 + 5x + 2$ ✓

b. $2x^2 + 5x - 2$

c. $3x^2 + 5x + 2$

d. $2x^2 + 3x + 4$

$f(g(x)) = 3g(x) - 2 = 6x^2 + 15x + 4 \Rightarrow 3g(x) = 6x^2 + 15x + 6 \Rightarrow g(x) = 2x^2 + 5x + 2$.

5 Let $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x - 2}$. What is the domain of $(f \circ g)(x)$?

a. $(2, \infty)$ ✓

b. $[2, \infty)$

c. $[0, \infty)$

d. $x \neq 2$

For $g(x)$ to exist, $x \geq 2$. For $f(g(x)) = \frac{1}{\sqrt{x-2}}$ to exist, the denominator cannot be zero, requiring $x > 2$.

6 If $h(x) = (2x^2 - 1)^5$, identify inner function $g(x)$ and outer function $f(u)$ such that $h(x) = f(g(x))$.

a. $g(x) = 2x^2 - 1$ **and** $f(u) = u^5$ ✓

b. $g(x) = x^5$ and $f(u) = 2u^2 - 1$

c. $g(x) = 2x$ and $f(u) = (u - 1)^5$

d. $g(x) = 2x^2$ and $f(u) = u^5 - 1$

Direct decomposition gives inner $g(x) = 2x^2 - 1$ and outer power function $f(u) = u^5$.

Section 3: Logarithmic Laws & Base Changes

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7 Solve the logarithmic equation: $\log_4(x+3) + \log_4(x-3) = 2$.

a. 5 ✓

b. -5

c. 5 and -5

d. $\sqrt{19}$

Combine logs: $\log_4(x^2-9) = 2 \Rightarrow x^2 - 9 = 4^2 = 16 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$. Since $x > 3$, only $x = 5$ is valid.

8 Using the change of base formula, evaluate $\log_8(32)$.

a. $\frac{5}{3}$ ✓

b. $\frac{3}{5}$

c. 4

d. $\frac{4}{3}$

Express with base 2: $\log_8(32) = \frac{\log_2 32}{\log_2 8} = \frac{5}{3}$.

9 If $\ln x + \ln y = a$ and $\ln x - \ln y = b$, what is x?

a. $e^{(a+b)/2}$ ✓

b. e^{a+b}

c. e^{ab}

d. $\frac{e^a + e^b}{2}$

Add both equations: $2\ln x = a + b \Rightarrow \ln x = \frac{a+b}{2} \Rightarrow x = e^{(a+b)/2}$.

Section 4: Exponential Models & Logarithmic Graphs

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10 What is the vertical asymptote of the graph of $f(x) = 3\ln(2x - 8) + 1$?

a. $x = 4$ ✓

b. $x = 8$

c. $x = 0$

d. $y = 1$

The argument of the natural log equals zero at the vertical asymptote: $2x - 8 = 0 \Rightarrow 2x = 8 \Rightarrow x = 4$.

11 Solve the exponential equation for t: $500e^{0.04t} = 2000$.

a. $25\ln 4$

b. $\frac{\ln 4}{0.04}$

c. $50\ln 2$

d. **All of the above** ✓

Divide by 500: $e^{0.04t} = 4 \Rightarrow 0.04t = \ln 4 \Rightarrow t = \frac{\ln 4}{0.04} = 25\ln 4 = 50\ln 2$. All choices are algebraically equivalent.

12 What is the x-intercept of the graph of $g(x) = \log_2(x - 3) - 2$?

a. 7 ✓

b. 5

c. 4

d. 11

Set $g(x) = 0$: $\log_2(x - 3) = 2 \Rightarrow x - 3 = 2^2 = 4 \Rightarrow x = 7$.

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